

INTERNSHIP AND MASTER'S THESIS

Soaring flight Research directions

Stochastic motion, collective flight
and possible extensions to optimal control

Matteo Di Sante

October 2026 – February 2027

1 Data download and preprocessing: completed

Possible simplification of preprocessing, with checks on phase labels and measured observables.

2 Segmentation

Continuous observations
Validated phase labels



3 Stochastic characterisation

A common set of empirical tests for all later work



4 Stochastic model

Intermittency and memory

5 Solo vs group

Individual and joint motion

6 Exploratory extension: thermal statistics and stochastic optimal control

Parallel throughout: maintain and improve the graphical analysis tool.

Vilpellet's current implementation

Three binary indicators, eight possible emissions. Windows: 6, 30 and 90 fixes. Angular thresholds and transitions are per fix.

Calibrated near 1 Hz. The current port requires mean $\Delta t < 1.2$ s. Other cadences need adaptation and validation.

12–25 October [2 weeks]

Develop the existing Gaussian HMM prototype: continuous v_z , horizontal speed and turning features, measured over windows in seconds.

Discrete hidden phases remain. Validate labels, boundaries and durations on independent annotated flights and across cadences.

Time currently excluded from phase analysis

30.1% paragliders

70.9% hang gliders

26 October – 15 November [3 weeks] Main established focus: MSD and its effective exponent

Geometry

Anisotropy and directional covariance. Dependence on altitude, terrain, circuit type and equipment.

Scaling laws

$\langle |\Delta \mathbf{r}(\tau)|^q \rangle \sim \tau^{\zeta(q)}$: is $\zeta(q) = qH$ (monofractal)? Test distribution collapse, tails and temporal joint laws.

Memory

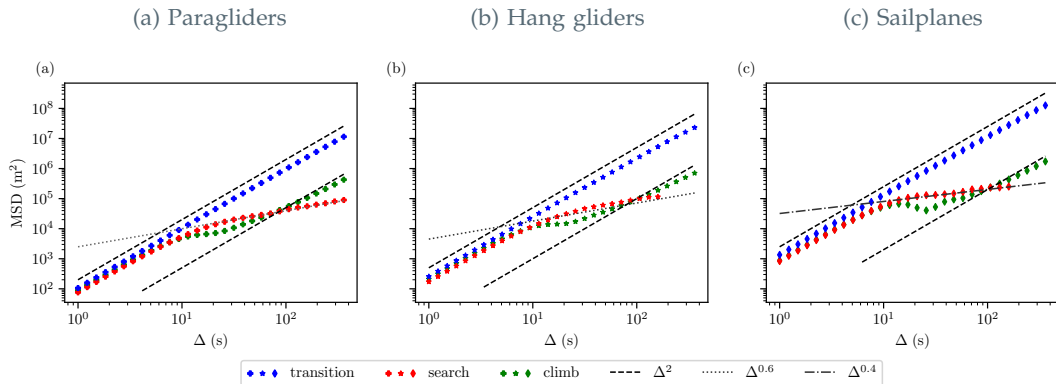
Velocity and heading correlations, phase durations, stationary increments, time vs ensemble averages and ageing.

Deliverable: empirical tests that can accept or reject specific model properties.

Finite scaling ranges and uncertainty are part of each result. An MSD power law alone does not establish self-similarity.

Empirical MSD in the three flight phases

5 / 12



Gliding / transition

Search

Climb

Approximately ballistic

Slow horizontal spreading

Circling with slow drift

Empirical reference: Vilpelle, Darmon & Benzaquen, [arXiv:2601.01293v1](https://arxiv.org/abs/2601.01293v1), Fig. 3. Separate cohort from our archive.

A two-state model with angular memory

6 / 12

16–30 November [about 2 weeks]

Simplification of the earlier model developed in Pisa

Gliding

$$\dot{\mathbf{r}} = v(t)(\cos \theta, \sin \theta)$$



Waiting

Search + climb

$$\dot{\mathbf{r}} \simeq 0 \text{ at coarse scales}$$

Empirical inputs

Glide/wait durations, speeds and

$$C_\theta(\tau) = \langle \cos[\theta(t + \tau) - \theta(t)] \rangle.$$

Test memory within and across glides.

Model test

Choose angular dynamics from the data.

Can the model reproduce the full characterisation, including long-time transport?

Solo flight and sustained group flight

7 / 12

30 November – 23 December [about 3 weeks + handover]

Candidate flights

Nearby launches in a common time window.
Current viewer functionality.



Sustained association

Simultaneous proximity, duration and co-motion.
Shared climb episodes.



Comparison

Reuse the stochastic tests from stage 3.

Validation

Vary distance, altitude and duration thresholds.
Compare with suitable surrogate pairs and manually inspect borderline cases.

Controls

Match site/day, circuit, altitude and equipment.
Treat missing nearby flights as incomplete observation of possible companions.

A common departure window identifies candidates. A group may form or split during flight, so labels can vary in time.

An observable starting point

A multi-year spatial density of recorded climb locations, conditional on aircraft visits and the phase detector.

Estimate exposure and uncertainty. Separate repeated visits, climb events and distinct flights. Compare specific sites and terrain features.

$$\dot{z}_{\text{glider}} = w_{\text{air}} - s_{\text{sink}}$$

A normalized map of observed climbs describes sampled activity. A physical occurrence probability requires a sampling model.

Quantities still to identify

Thermal encounter probability for a defined area and interval. Strength distribution given a location, altitude and conditions.

The archive alone does not identify a complete instantaneous thermal field. Similar terrain classes can hide different local structure.

Observed climb rate is a proxy. Estimating air updraft speed needs a sink model and relevant flight conditions.

4 January – 7 February 2027 [5 weeks] Scope to refine after reading and a feasibility check

Starting reference

Almgren & Tourin study optimal speed and climb/cruise decisions under uncertain atmospheric conditions using HJB equations.

Possible extension: horizontal route choice in (x, y) , retaining altitude z and the information available to the pilot.

A possible pilot study

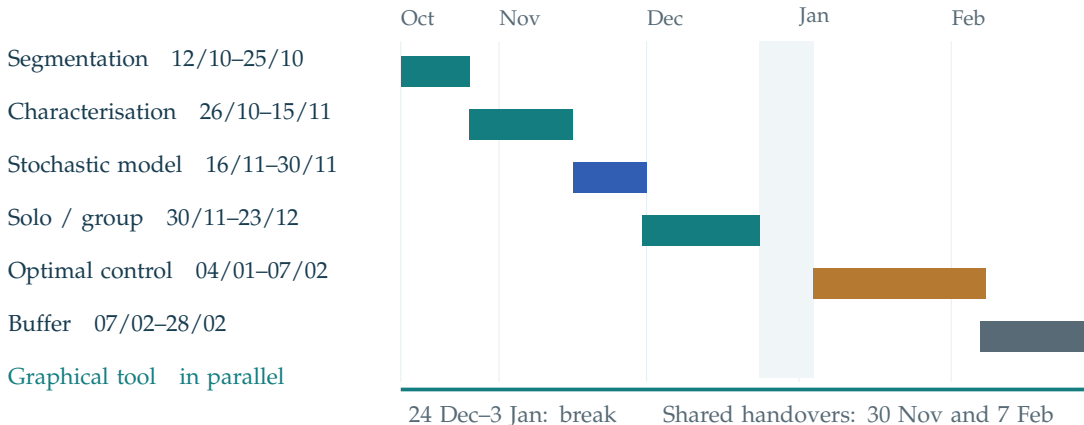
1. Reproduce a simple reference problem.
2. Choose the state, controls and objective.
3. Add an empirical environmental model.
4. Compare policies on held-out flights.

Candidate loss: expected travel time + penalty for landing short, with a specified information set.

Almgren & Tourin, *Optimal Control Applications and Methods* 36, 475–495 (2015). A thermal map alone is insufficient.

October 2026 – February 2027

Calendar windows with approximate effort estimates.



Buffer: delays or deeper work on group flight, the stochastic model or optimal control, guided by the results.

One practical interface for inspection, validation and comparison

Easier to run

Documented installation, portable paths, a small example dataset and checks on another computer.

Easier to use

Clear controls and defaults, visible units, reliable loading and fewer manual setup steps.

Research needs

Review phase labels, inspect group membership and compare thermal maps and stochastic observables.

Reproducible exports should retain the cohort, filters and analysis settings used to obtain each figure.

Continuous maintenance and targeted additions according to the needs of stages 2–6.

Shared foundation

Validated segmentation and a defensible stochastic characterisation.

Review at the end of October and in mid-November.

Main research paths

An empirical test of the intermittent model.

A robust definition and comparison of solo/group motion.

Conditional extension

Thermal inference and a limited optimal-control study.

Feasibility decision before the winter break.

February remains flexible: deepen the strongest result and address delays before the end of the internship.

Sampling restriction: coverage and denominators

A1

Archived cleaned support, segmentation export of 16 September 2026. Figures rechecked on 7 October.

	Paragliders	Hang gliders
Flights in the coverage archive	156,406	6,094
Flights with any classified segment	104,636 (66.9%)	1,670 (27.4%)
Supported time at eligible cadence	70.7%	29.2%
Supported time classified, all gates	69.9%	29.1%
Fixes at eligible cadence	89.3%	62.0%
Fixes classified, all gates	88.2%	61.7%

Eligible cadence: mean $\Delta t < 1.2$ s. Other gates: 3,600 fixes per flight, at least 30 per segment, valid increasing timestamps.

Time = sum of segment end minus start, excluding gaps. Counts concern these archived cleaned segments, not all downloaded tracks.

For stationary, zero-mean velocity with finite second moment:

$$M(\tau) = 2 \int_0^\tau (\tau - s) C_v(s) ds$$

$C_v(s) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t+s) \rangle$. With drift, apply this relation to centred displacement and covariance.

Finite memory

If $\int_0^\infty C_v(s) ds$ is finite and positive, $M(\tau) \sim \tau$.
Constant-speed angular Brownian motion is a baseline example.

Persistent tail

If $C_v(s) \sim As^{-\gamma}$, $A > 0$ and $0 < \gamma < 1$, then $M(\tau) \sim \tau^{2-\gamma}$. Test whether the model supports this limit.